

Effect of adding various powders with small particles on the magnetic properties and core-loss reduction of soft magnetic composites

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Introduction

Motivation

- ✓ Soft magnetic composites (SMCs) with high saturation magnetization (M_s), low coercivity (H_c), and low core-loss are essential for power-inductor to achieve high energy efficiency at high frequencies.
- ✓ Especially, high magnetic permeability is required for high inductance.
- ✓ Improving packing fraction of the composite by applying small particles into bigger particles can lead to improvement of magnetic permeability.
- ✓ We made toroidal core type samples adding CIP and Fe-50Ni powder into Fe-Si-Cr powders, and measured permeability and core-loss to check if the core is suitable for high-frequency application.



Introduction

Permeability

$$L = \mu_r \mu_0 \frac{AN^2}{l}$$

L : inductance, N : number of windings of the coil, A : cross-sectional area of the core, l : magnetic field, μ_r : relative permeability, μ_0 : permeability in a vacuum

High permeability is required for high inductance.

Ollendorff's equation

$$\mu = \mu_0 + \frac{\eta \mu_0 (\mu_m - \mu_0)}{N(1 - \eta)(\mu_m - \mu_0) + \mu_0}$$

η : volume packing fraction, N : coefficient of the demagnetizing field, μ_m : permeability of the material

Improving packing fraction of the core can improve permeability.



Introduction

Sample Preparation

Sieve Fe-Si-Cr powders into under 25 μm particle sizes



Mix Fe-Si-Cr powder with 10, 20, 30, 40 wt% of CIP and Fe-50Ni powders



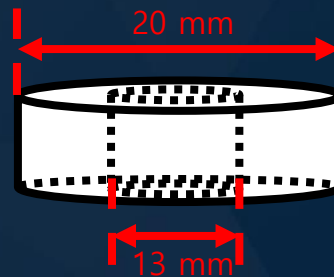
Add 3 wt% of epoxy resin into the mixed powder and mix all together



Put the mixed powder in a toroidal core molder and press it in 2 tons



Dry the fabricated toroidal cores in 200 $^{\circ}\text{C}$ for two hours

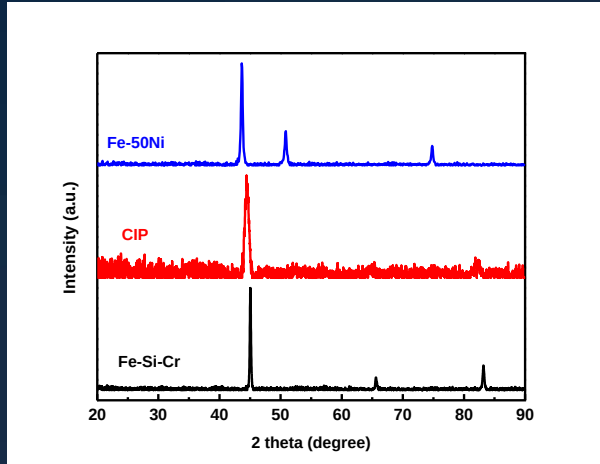


Toroidal core



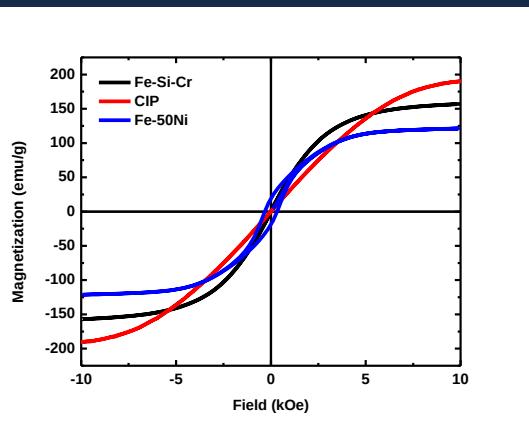
Experiments & Results

X-ray Diffraction (XRD)



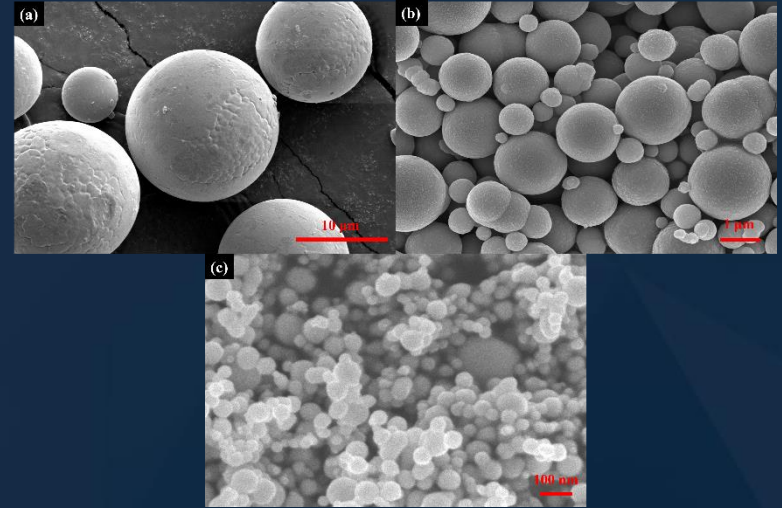
- ✓ Fe-Si-Cr, CIP – BCC structure
- ✓ Fe-50Ni – FCC structure

Vibrating Sample Magnetometer (VSM)



- ✓ All powders show good soft magnetic properties with high M_s and low H_c
- ✓ Fe-50Ni shows slightly high H_c – act as superparamagnetic because they are nanoparticles.

Scanning Electron Microscope (SEM)

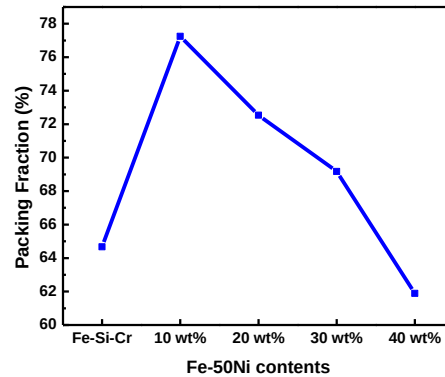
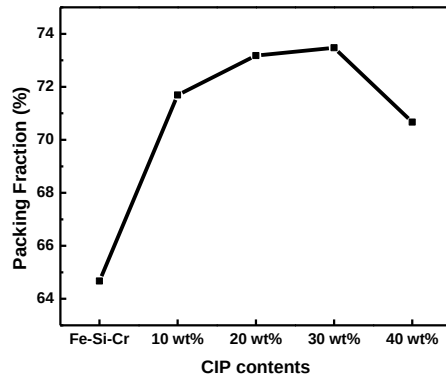
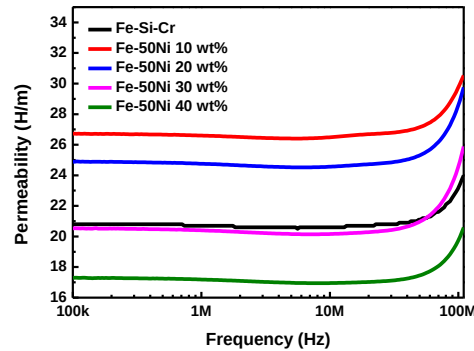
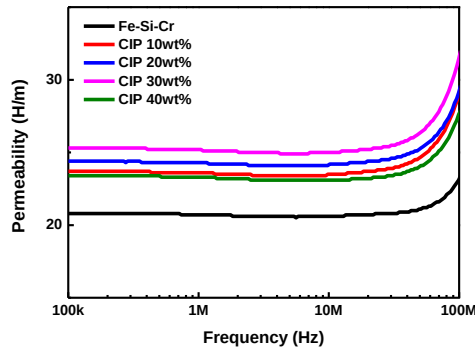


- ✓ Powder morphologies of (a) Fe-Si-Cr, (b) CIP, and (c) Fe-50Ni.
- ✓ All samples showed spherical shape.
- ✓ Particle size Fe-Si-Cr > CIP > Fe-50Ni



Experiments & Results

Permeability & Density

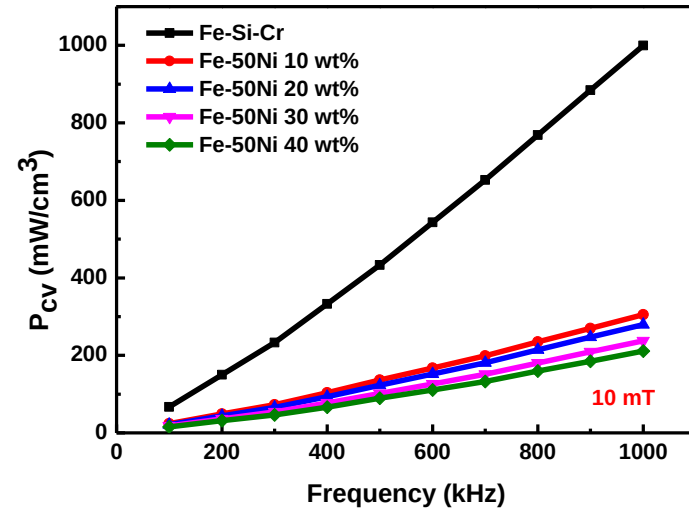
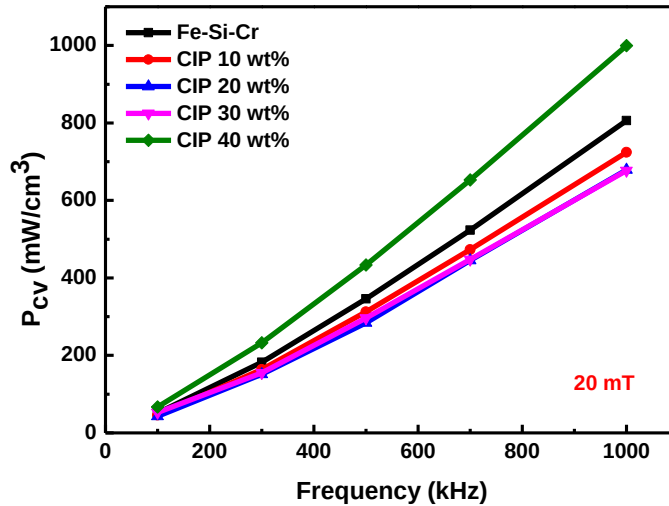


- ✓ Permeability vs. packing fraction – similar tendency.
- ✓ Addition of small powders → increase of packing fraction.
- ✓ For Fe-50Ni addition, containing 40 wt% of Fe-50Ni showed a reduced packing fraction and permeability.
- ✓ Showing the relationship with Ollendorff's equation



Experiments & Results

Core-loss



- ✓ Addition of CIP was conducted under $B_m = 20\text{mT}$, and Fe-50Ni under $B_m = 10\text{ mT}$.
- ✓ Addition of small powders showed reduction of core-loss.
- ✓ Especially, addition of 10 wt% of Fe-50Ni showed a significant decrease of core-loss compared to pure Fe-Si-Cr core.



Conclusions

- ✓ In this study, the **permeability, and core-loss** dependency of various cores with addition of various small particles on Fe-Si-Cr based cores, were investigated.
- ✓ CIP and Fe-50Ni addition showed a **change of packing fraction** of cores and this led to **improvement of permeability**.
- ✓ Addition of **30 wt% CIP** showed the highest packing fraction and permeability among CIP added cores, and addition of **10 wt% Fe-50Ni** showed the same tendency.
- ✓ Cores showing highest packing fraction and permeability also showed the lowest core-loss. This is because of the **reduction of air gap between particles**, applied by improvement of packing fraction.